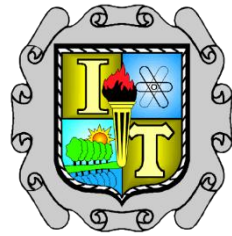




**TECNOLÓGICO
NACIONAL DE MÉXICO**



Tecnológico Nacional de México

Instituto Tecnológico de Saltillo

Ingeniería en Sistemas Computacionales

Cómputo en la Nube

Unidad 5

5.8 – Instalación de entorno LAMP

Alumna:

Silvia Guadalupe Ramos Mendoza 22050726

Docente: Ing. Miguel Salazar del Bosque

Saltillo, Coahuila; a 28 de abril de 2026

Introducción

El servidor web LAMP (Linux, Apache, MySQL, PHP) es una de las configuraciones más utilizadas en el desarrollo web, ya que proporciona un entorno completo para alojar y ejecutar aplicaciones del lado del servidor. En esta actividad se llevó a cabo la instalación y configuración de Apache, PHP y MySQL sobre la máquina virtual Ubuntu Server creada previamente en Microsoft Azure, con el objetivo de poner en marcha un servidor web funcional y verificar el correcto funcionamiento de cada componente.

Conectarse por SSH

```
silvia@mv-silvia: ~  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1011-azure aarch64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Wed Apr 29 01:03:32 UTC 2026  
  
System load:  0.06      Processes:      140  
Usage of /:   6.1% of 28.02GB  Users logged in:  0  
Memory usage: 30%      IPv4 address for eth0: 10.0.0.4  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Sat Apr 25 04:29:44 2026 from 177.227.194.160  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
silvia@mv-silvia:~$
```

Actualizar Sistema

```
silvia@mv-silvia: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
silvia@mv-silvia:~$ sudo apt update && sudo apt upgrade -y  
Hit:1 http://ports.ubuntu.com/ubuntu-ports noble InRelease  
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates InRelease [126 kB]  
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-backports InRelease [126 kB]  
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-security InRelease [126 kB]  
Get:5 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 Packages [15.3 MB]  
Get:6 http://ports.ubuntu.com/ubuntu-ports noble/universe Translation-en [5982 kB]  
Get:7 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 Components [3573 kB]  
Get:8 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 c-n-f Metadata [295 kB]  
Get:9 http://ports.ubuntu.com/ubuntu-ports noble/multiverse arm64 Packages [223 kB]  
Get:10 http://ports.ubuntu.com/ubuntu-ports noble/multiverse Translation-en [118 kB]  
Get:11 http://ports.ubuntu.com/ubuntu-ports noble/multiverse arm64 Components [31.6 kB]  
Get:12 http://ports.ubuntu.com/ubuntu-ports noble/multiverse arm64 c-n-f Metadata [7152 B]  
Get:13 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 Packages [2044 kB]  
Get:14 http://ports.ubuntu.com/ubuntu-ports noble-updates/main Translation-en [346 kB]  
Get:15 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 Components [174 kB]  
Get:16 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 c-n-f Metadata [16.8 kB]  
Get:17 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 Packages [1664 kB]  
Get:18 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe Translation-en [324 kB]  
Get:19 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 Components [385 kB]  
Get:20 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 c-n-f Metadata [33.1 kB]  
Get:21 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted arm64 Packages [4370 kB]  
Get:22 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted Translation-en [702 kB]  
Get:23 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted arm64 Components [212 B]  
Get:24 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted arm64 c-n-f Metadata [420 B]  
Get:25 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse arm64 Packages [52.3 kB]  
Get:26 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse Translation-en [10.2 kB]
```

```
silvia@mv-silvia: ~  
Processing triggers for initramfs-tools (0.142ubuntu25.8) ...  
update-initramfs: Generating /boot/initrd.img-6.17.0-1011-azure  
W: Possible missing firmware /lib/firmware/renesas_usb_fw.mem for built-in driver xhci_pci_renesas  
System running in EFI mode, skipping.  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
Restarting services...  
/etc/needrestart/restart.d/systemd-manager  
systemctl restart chrony.service multipathd.service polkit.service rsyslog.service ssh.service systemd-journald.service  
systemd-networkd.service systemd-resolved.service systemd-udevd.service udisks2.service walinuxagent.service  
  
Service restarts being deferred:  
systemctl restart ModemManager.service  
/etc/needrestart/restart.d/dbus.service  
systemctl restart networkd-dispatcher.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
silvia @ session #1: apt[1525], sshd[1050]  
silvia @ user manager service: systemd[1055]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
silvia@mv-silvia:~$
```

Instalar Apache

```
silvia@mv-silvia: ~  
silvia @ session #1: apt[1525], sshd[1050]  
silvia @ user manager service: systemd[1055]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
silvia@mv-silvia:~$ sudo apt install apache2 -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64  
  liblua5.4-0 ssl-cert  
Suggested packages:  
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser  
The following NEW packages will be installed:  
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64  
  liblua5.4-0 ssl-cert  
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.  
Need to get 2071 kB of archives.  
After this operation, 14.0 MB of additional disk space will be used.  
Get:1 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libapr1t64 arm64 1.7.2-3.1ubuntu0.1 [106 kB]  
Get:2 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libaprutil1t64 arm64 1.6.3-1.1ubuntu7 [93.9 kB]  
Get:3 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libaprutil1-dbd-sqlite3 arm64 1.6.3-1.1ubuntu7 [11.2 kB]  
Get:4 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libaprutil1-ldap arm64 1.6.3-1.1ubuntu7 [9046 B]  
Get:5 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 liblua5.4-0 arm64 5.4.6-3build2 [158 kB]  
Get:6 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 apache2-bin arm64 2.4.58-1ubuntu8.11 [1324 kB]  
Get:7 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 apache2-data all 2.4.58-1ubuntu8.11 [163 kB]  
Get:8 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 apache2-utils arm64 2.4.58-1ubuntu8.11 [97.4 kB]  
Get:9 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 apache2 arm64 2.4.58-1ubuntu8.11 [90.2 kB]  
Get:10 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]  
Fetched 2071 kB in 2s (1050 kB/s)
```

```

silvia@mv-silvia: ~
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
silvia @ session #1: sshd[1050]
silvia @ user manager service: systemd[1055]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
silvia@mv-silvia:~$

```

```

silvia@mv-silvia: ~
Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
silvia @ session #1: sshd[1050]
silvia @ user manager service: systemd[1055]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
silvia@mv-silvia:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 01:06:25 UTC; 53s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 7190 (apache2)
    Tasks: 55 (limit: 1089)
   Memory: 5.0M (peak: 5.5M)
      CPU: 28ms
   CGroup: /system.slice/apache2.service
           └─7190 /usr/sbin/apache2 -k start
             └─7193 /usr/sbin/apache2 -k start
               └─7194 /usr/sbin/apache2 -k start

Apr 29 01:06:25 mv-silvia systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 29 01:06:25 mv-silvia systemd[1]: Started apache2.service - The Apache HTTP Server.
silvia@mv-silvia:~$

```

Instalar PHP

```
silvia@mv-silvia: ~  
Apr 29 01:06:25 mv-silvia systemd[1]: Started apache2.service - The Apache HTTP Server.  
silvia@mv-silvia:~$ sudo apt install php libapache2-mod-php -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline  
Suggested packages:  
  php-pear  
The following NEW packages will be installed:  
  libapache2-mod-php libapache2-mod-php8.3 php php-common php8.3 php8.3-cli php8.3-common php8.3-opcache  
  php8.3-readline  
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.  
Need to get 5003 kB of archives.  
After this operation, 24.1 MB of additional disk space will be used.  
Get:1 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 php-common all 2:93ubuntu2 [13.9 kB]  
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 php8.3-common arm64 8.3.6-0ubuntu0.24.04.8 [738 kB]  
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 php8.3-opcache arm64 8.3.6-0ubuntu0.24.04.8 [580 kB]  
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 php8.3-readline arm64 8.3.6-0ubuntu0.24.04.8 [13.2 kB]  
Get:5 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 php8.3-cli arm64 8.3.6-0ubuntu0.24.04.8 [1855 kB]  
Get:6 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libapache2-mod-php8.3 arm64 8.3.6-0ubuntu0.24.04.8 [1785 kB]  
Get:7 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libapache2-mod-php all 2:8.3+93ubuntu2 [4224 B]  
Get:8 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 php8.3 all 8.3.6-0ubuntu0.24.04.8 [9170 B]  
Get:9 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 php all 2:8.3+93ubuntu2 [4076 B]  
Fetched 5003 kB in 1s (6147 kB/s)  
Selecting previously unselected package php-common.  
(Reading database ... 75597 files and directories currently installed.)  
Preparing to unpack .../0-php-common_2%3a93ubuntu2_all.deb ...
```

```
silvia@mv-silvia: ~  
apache2_switch_mpm Switch to prefork  
apache2_invoke: Enable module php8.3  
Setting up php8.3 (8.3.6-0ubuntu0.24.04.8) ...  
Setting up libapache2-mod-php (2:8.3+93ubuntu2) ...  
Setting up php (2:8.3+93ubuntu2) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for php8.3-cli (8.3.6-0ubuntu0.24.04.8) ...  
Processing triggers for libapache2-mod-php8.3 (8.3.6-0ubuntu0.24.04.8) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
Restarting services...  
  
Service restarts being deferred:  
  /etc/needrestart/restart.d/dbus.service  
  systemctl restart networkd-dispatcher.service  
  systemctl restart systemd-logind.service  
  systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
  silvia @ session #1: sshd[1050]  
  silvia @ user manager service: systemd[1055]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
silvia@mv-silvia:~$
```

Instalar MySQL

```
silvia@mv-silvia: ~  
silvia@mv-silvia:~$ sudo apt install mysql-server -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin  
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl  
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl  
  liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common  
  mysql-server-8.0 mysql-server-core-8.0  
Suggested packages:  
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl  
  libwww-perl mailx tinymc  
The following NEW packages will be installed:  
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin  
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl  
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl  
  liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common  
  mysql-server mysql-server-8.0 mysql-server-core-8.0  
0 upgraded, 28 newly installed, 0 to remove and 0 not upgraded.  
Need to get 29.3 MB of archives.  
After this operation, 243 MB of additional disk space will be used.  
Get:1 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 mysql-common all 5.8+1.1.0build1 [6746 B]  
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 mysql-client-core-8.0 arm64 8.0.45-0ubuntu0.24.04.1  
[2807 kB]  
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 mysql-client-8.0 arm64 8.0.45-0ubuntu0.24.04.1 [22.5  
kB]  
Get:4 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libevent-pthreads-2.1-7t64 arm64 2.1.12-stable-9ubuntu2 [793  
2 B]  
Get:5 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libmecab2 arm64 0.996-14ubuntu4 [188 kB]
```

```
silvia@mv-silvia: ~  
Setting up libhtml-parser-perl:arm64 (3.81-1build3) ...  
Setting up libhttp-message-perl (6.45-1ubuntu1) ...  
Setting up mysql-server (8.0.45-0ubuntu0.24.04.1) ...  
Setting up libcgi-pm-perl (4.63-1) ...  
Setting up libhtml-template-perl (2.97-2) ...  
Setting up libcgi-fast-perl (1:2.17-1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
Restarting services...  
  
Service restarts being deferred:  
  /etc/needrestart/restart.d/dbus.service  
systemctl restart networkd-dispatcher.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
  silvia @ session #1: sshd[1050]  
  silvia @ user manager service: systemd[1055]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
silvia@mv-silvia:~$
```

```

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
silvia @ session #1: sshd[1050]
silvia @ user manager service: systemd[1055]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
silvia@mv-silvia:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 01:09:39 UTC; 1min 33s ago
   Process: 14762 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
  Main PID: 14788 (mysqld)
    Status: "Server is operational"
     Tasks: 37 (limit: 1089)
  Memory: 359.5M (peak: 380.0M)
     CPU: 724ms
    CGroup: /system.slice/mysql.service
            └─14788 /usr/sbin/mysqld

Apr 29 01:09:38 mv-silvia systemd[1]: Starting mysql.service - MySQL Community Server...
Apr 29 01:09:39 mv-silvia systemd[1]: Started mysql.service - MySQL Community Server.
silvia@mv-silvia:~$

```

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
silvia@mv-silvia:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 01:09:39 UTC; 1min 33s ago
   Process: 14762 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
  Main PID: 14788 (mysqld)
    Status: "Server is operational"
     Tasks: 37 (limit: 1089)
  Memory: 359.5M (peak: 380.0M)
     CPU: 724ms
    CGroup: /system.slice/mysql.service
            └─14788 /usr/sbin/mysqld

Apr 29 01:09:38 mv-silvia systemd[1]: Starting mysql.service - MySQL Community Server...
Apr 29 01:09:39 mv-silvia systemd[1]: Started mysql.service - MySQL Community Server.
silvia@mv-silvia:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>

```


Probar Servidor Web



The screenshot shows a web browser window with the address bar displaying "No seguro 9.234.40.231". The browser tabs include "GPSen2026: Control Integrado", "(21) WhatsApp", "Discord | @Hugo", "Infraestructura de proceso - M...", and "Apache2 Ubuntu Default Page". The page content features the Ubuntu logo, the title "Apache2 Default Page", and a green button that says "It works!". Below this, there is a paragraph explaining that this is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It mentions that the page is based on the equivalent page on Debian and that the Apache HTTP server installed at this site is working properly. It advises replacing the file located at `/var/www/html/index.html` before continuing to operate the HTTP server. A section titled "Configuration Overview" follows, stating that Ubuntu's Apache2 default configuration is different from the upstream default configuration and is split into several files optimized for interaction with Ubuntu tools. It notes that the configuration system is fully documented in `/usr/share/doc/apache2/README.Debian.gz` and refers to the full documentation for the web server itself, which can be found by accessing the manual if the `apache2-doc` package was installed on this server. The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

Conclusión

La instalación del servidor web resultó ser una actividad sencilla, ya que previamente se había tenido contacto con este proceso, lo que permitió ejecutar cada paso con mayor fluidez y seguridad. Se logró instalar y verificar el correcto funcionamiento de Apache, PHP y MySQL, comprobando que el servidor respondía correctamente tanto desde la terminal como desde el navegador. Esta práctica reforzó los conocimientos adquiridos sobre administración de servidores en entornos de nube.